INCH-POUND
MIL-F-24402/5A(SH)
10 February 1992
SUPERSEDING
MIL-F-24402/5
3 April 1990

MILITARY SPECIFICATION SHEET

FILTER INDICATORS, DIFFERENTIAL PRESSURE, HYDRAULIC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-24402.

1. Classification. Filter differential pressure indicators covered by this specification sheet shall be designated using part or identifying numbers (PIN) as follows:

M	24402	/5	-G150D
Prefix to indicate a military specification	Specification number	Applicable specification sheet	Dash number (see 2 and 3)

1.1 Dash number. The dash number for indicators consist of a letter designating the indicator type followed by a number representing the range or actuating pressure of the indicator and a letter suffix, when applicable, designating the flange configuration.

Examples: M24402/5-G150D is for a gauge-type indicator with a minimum range of 150 lb/in^2 differential and a dual bolt circle, eight hole mounting flange.

M24402/5-M45 is for a mechanical pop-up indicator with an actuation pressure of 45 lb/in^2 differential.

AMSC N/A
FSC 4330
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2. Indicator type.

2.1 Gauge indicators. Gauge indicators shall be designated by the following dash numbers:

-G150 Gauge shall have a minimum range of 30 to 150 lb/in^2 differential.

2.2 Mechanical indicators. Mechanical indicators shall be designated by the following dash numbers:

-M45 Actuation pressure shall be 45 ± 5 lb/in^2 differential
-M90 Actuation pressure shall be 90 ± 10 lb/in^2 differential

2.3 Electrical indicators. Electrical indicators with mechanical pop-up indicator shall be designated by the following dash numbers:

-E45 Actuation pressure shall be 45 ± 5 lb/in^2 differential
-E90 Actuation pressure shall be 90 ± 10 lb/in^2 differential

3. Mounting flange configuration. The differential pressure indicator mounting flange configuration shall be designated by a suffix as follows:

No suffix - Standard 4-hole flange to mate with figure 1 gland of MIL-F-24402.

D - Dual bolt circle, 8-hole flange to mate with figure 1 gland of MIL-F-24402 but also containing four holes of 0.172 +/- .003 inch diameter equally spaced on a 1.069 inch diameter bolt circle.

The D suffix is included in the part number to identify an indicator suitable for multiple mounting configurations for logistic support.

4. Part number cross reference. Table I provides a cross-reference between old (MIL-F-24402D and earlier) and new part numbers.

TABLE I. Cross reference of part numbers.

New part number	Old part numbers	
(Fatigue rated)	(Fatigue rated)	(Non-fatigue rated)
M24402/5-E90	M24402 form E-3	M24402 form E-0
M24402/5-E45	-	-
M24402/5-G150	M24402 form G-3	M24402 form G-0
M24402/5-M90	M24402 form M-3	M24402 form M-0
M24402/5-M45	-	-

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4.1 Substitution of assemblies. Indicators with new part numbers may be substituted for corresponding indicators with old part numbers as shown in table I. Only indicators with old part numbers that are fatigue rated may be substituted for indicators with new part numbers.

5. Requirements.

5.1 Configuration. The indicators shall comply with the envelope dimensions shown on figure 1.

(a) Male gland dimensions. The male gland shall comply with the dimensions shown on figure 2.
(b) Seals. The O-rings and backup ring specified on figure 2 shall be supplied already assembled onto the indicator.
(c) Mounting screws. When acquired separately from a filter housing, differential pressure indicators shall be provided with four mounting screws. The mounting screws shall be NAS 1351-4H*P where * is the length of the cap screw in sixteenths of an inch. Length of cap screw shall be selected to provide between 3/8 and 1/2 inch thread engagement. The dual bolt circle indicator with eight mounting holes shall be provided only with the mounting screws identified above. An instruction sheet shall be packaged with each indicator and shall contain as a minimum the following statement:

"Installation information. For installation instructions, fastener torques and use of mounting screws other than those supplied, see NAVSEA S9086-S4-STM-000, Naval Ships' Technical Manual Chapter 556 (Hydraulic Equipment), Section 7."

5.2 Maximum rated operating pressure. The indicators shall have a rated operating pressure of 3000 lb/in^2.

5.3 Surge lockout pressure. For mechanical and electrical indicators. The minimum surge lockout differential pressure shall be 15 lb/in^2 greater than the nominal actuation differential pressure.

Preparing activity:
Navy - SH
(Project 4330-N084)

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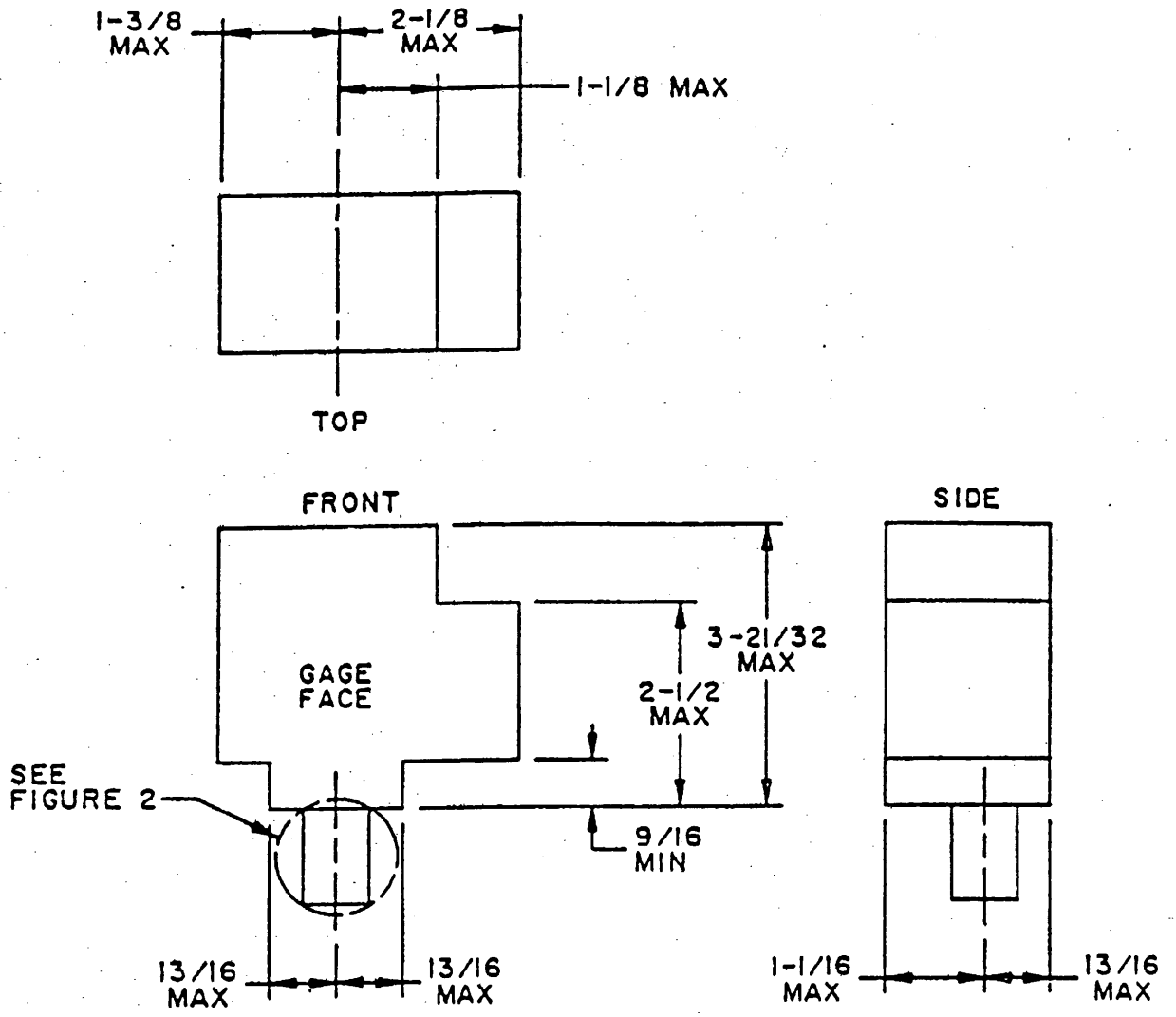
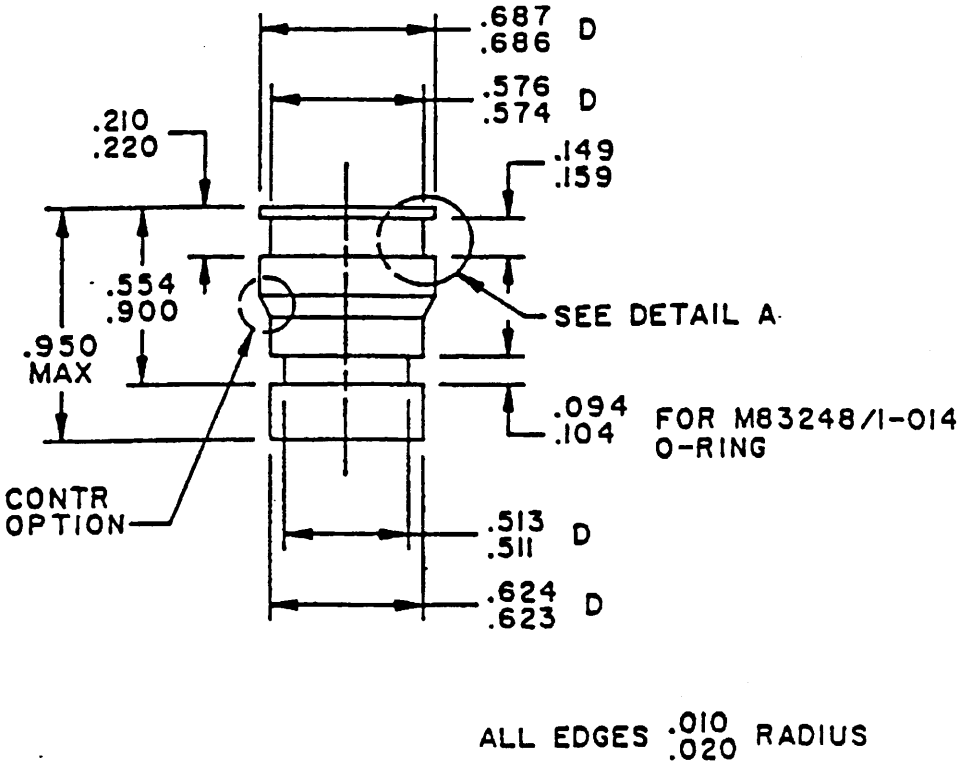
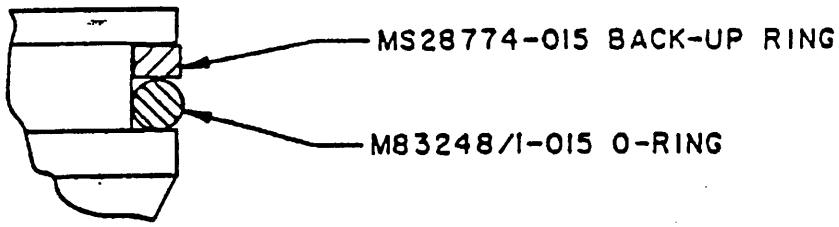


FIGURE 1. Pressure indicator envelope dimensions.

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DETAIL A

FIGURE 2. Male gland dimensions.

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